Introduction to Bayesian Data Analysis

Lecture 8: Further topics

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Further topics?

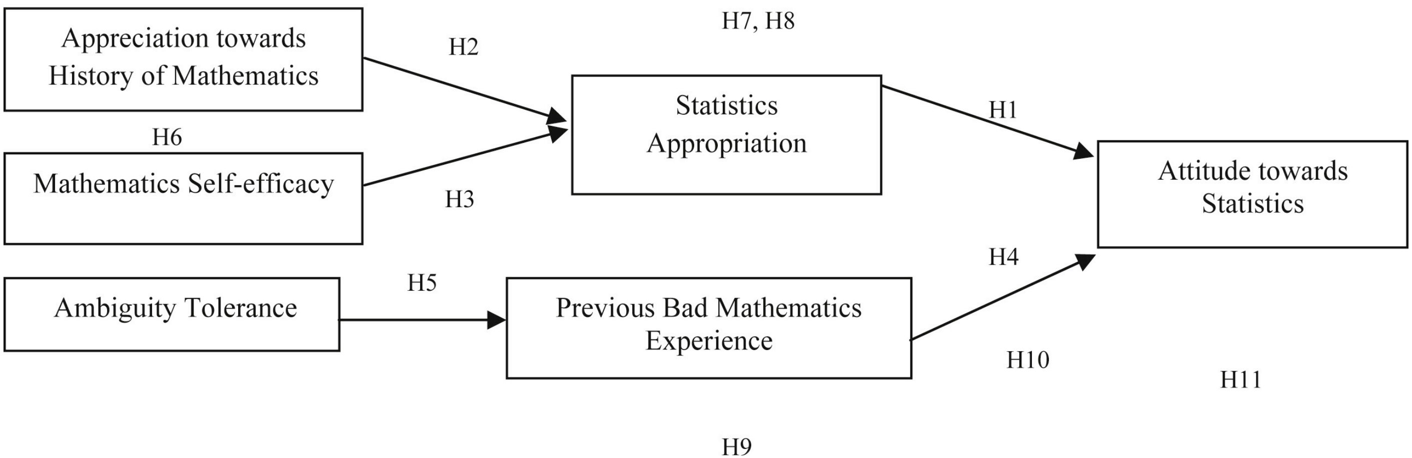
Overview

- Formal Cognitive Models
- Signal Detection Theory
- Drift-Diffusion Model
- Other Further Topics

Formal Cognitive Models

There are many things that people call models.

E.g. Prayoga, T., & Abraham, J. (2017). A psychological model explaining why we love or hate statistics.

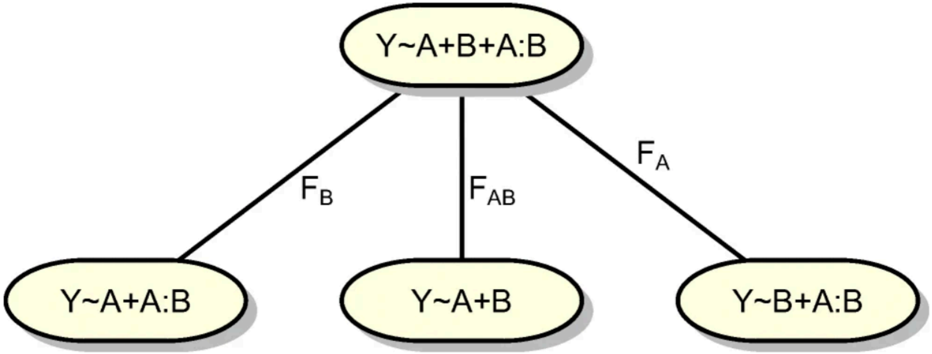


This is a verbal model.

Formal Cognitive Models

There are many things that people call models.

E.g. Rouder, Schnuerch, Haaf & Morey (2023). Principles of Model Specification in ANOVA Designs.



These are statistical models.

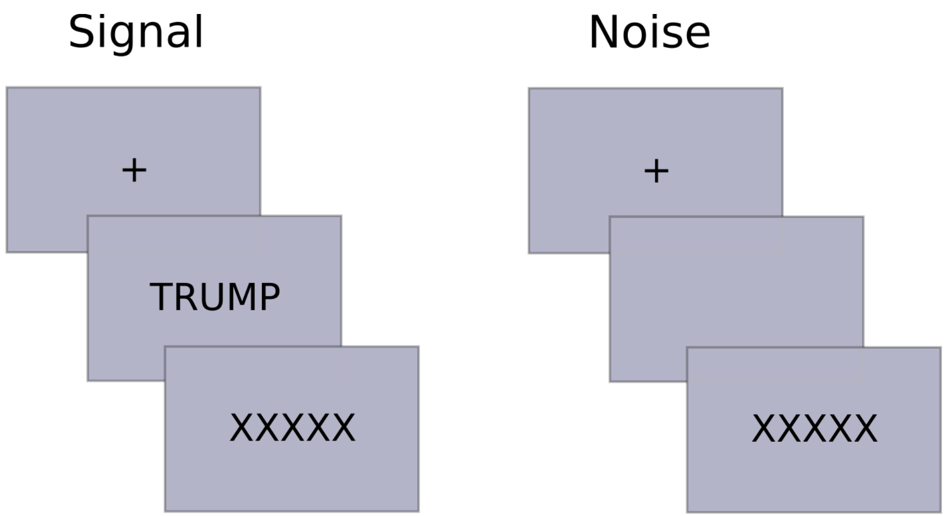
Formal Cognitive Models

> "A mathematical model or theory is a set of mathematical structures, including a set of *linkage statements*" (vanZandt & Townsend, 2012)

- Behavioral variables are related to components of cognitive processes using equations.
- Model parameters and function are interpretable as cognitive processes.
- As with statistical models, behavior needs to be quantifiable (e.g. accuracy, response time).

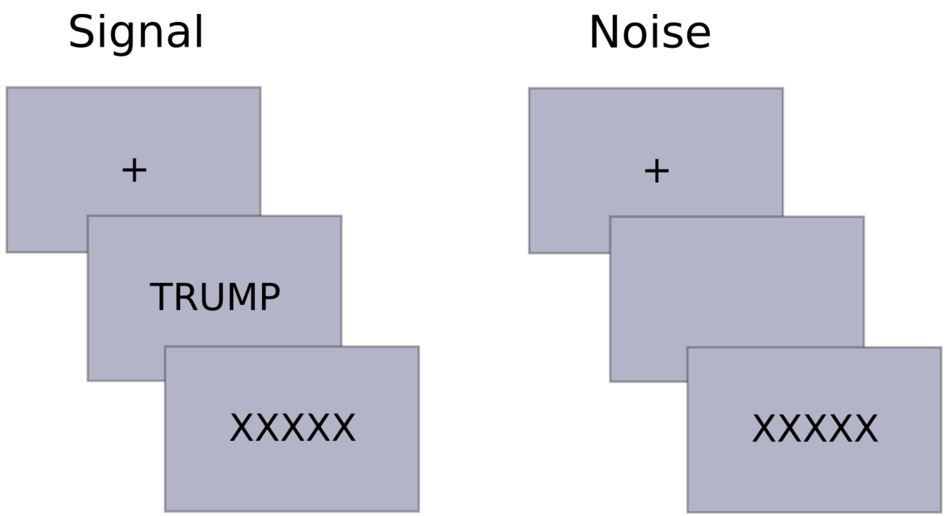
Signal Detection Theory

Signal detection experiment



Stimulus	Present response	Absent Response	Total
Signal	75	25	100
Noise	30	20	50
Total	105	45	

Signal detection experiment



Stimulus	Present response	Absent Response	Total
Signal	75 (Hits)	25 (Misses)	100
Noise	30 (False Alarms)	20 (Correct Rejections)	50
Total	105	45	

Signal Detection Theory vs. Signal detection experiment

- Signal Detection Theory (SDT): Model-based method of assessing performance
- Originally developed for signal detection experiments
- Now used for perception, memory, decision theory, attitude research, ...

- Originally for two-choice tasks
- Now extended to any form of forced-choice task including rating scales, continuous measures, ...

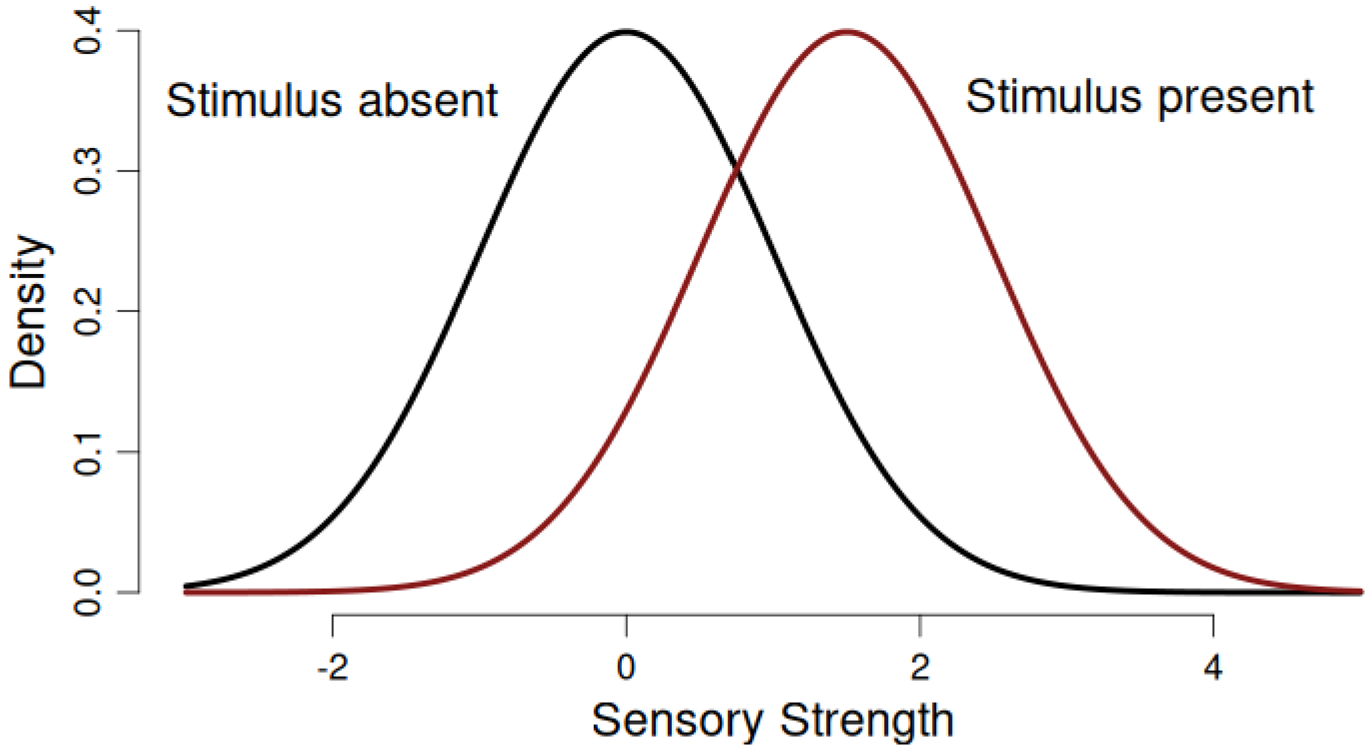
Green & Swets, 1966

Signal Detection Theory model

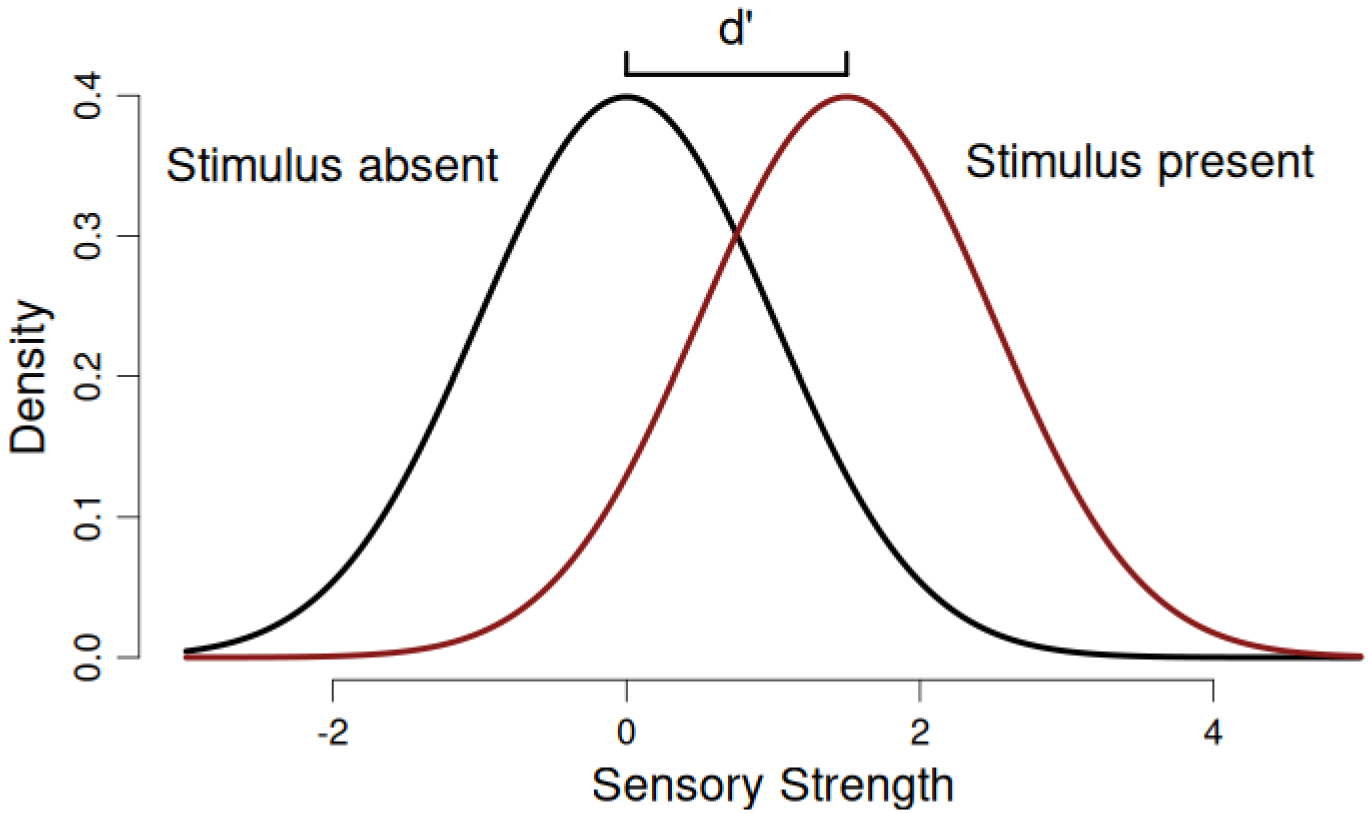
- General idea: Perception strength S varies gradually.
- On average, perceptual strength is higher when the stimulus is present compared to absent.

$$S \sim \begin{cases} \text{Normal}(\mu = d', \sigma^2 = 1), & \text{for signal-present trials,} \\ \text{Normal}(\mu = 0, \sigma^2 = 1), & \text{for signal-absent trials.} \end{cases}$$

SDT model

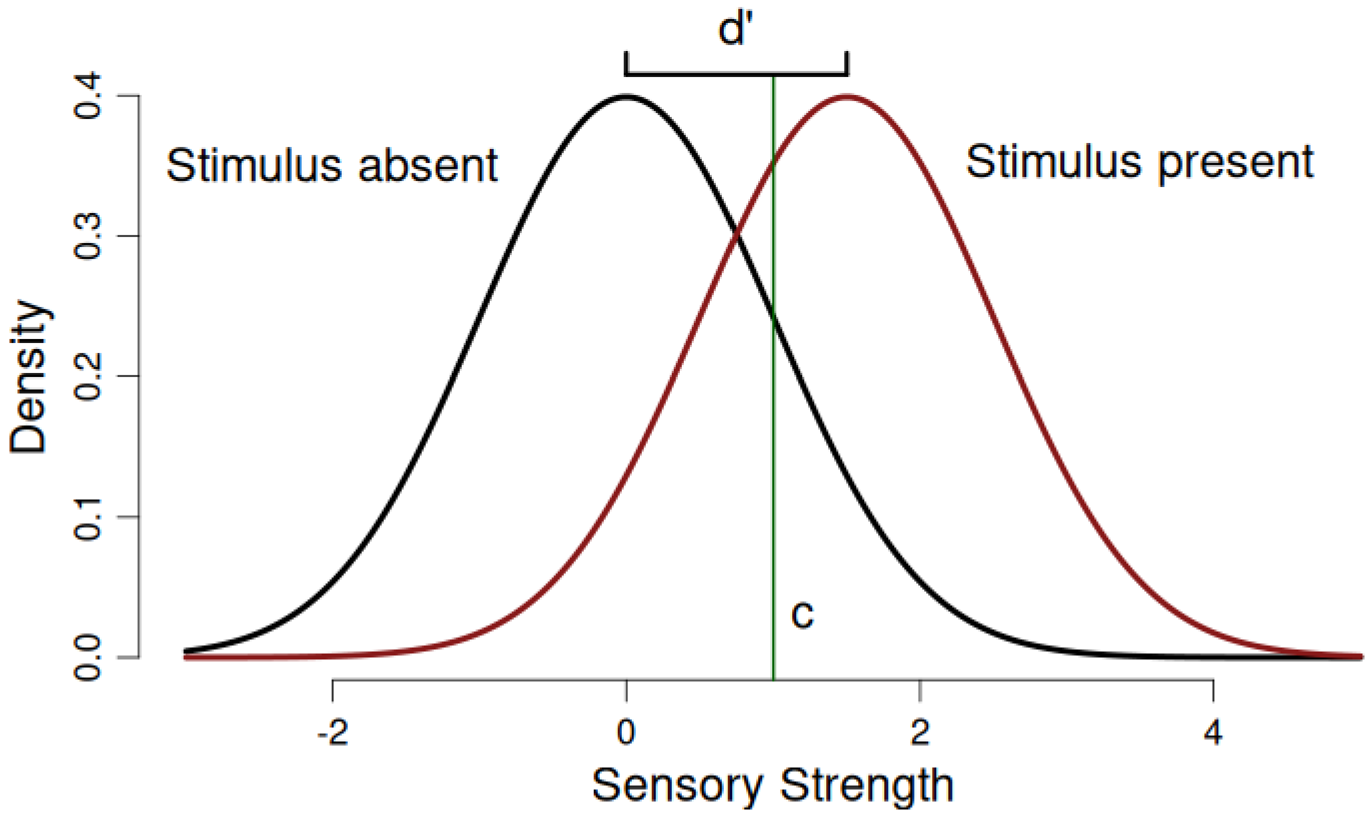

Stimulus absent
Stimulus present
Density
0.0
0.1
0.2
0.3
0.4
-2
0
2
4
Sensory Strength

SDT model



$d' =$ Sensitivity.

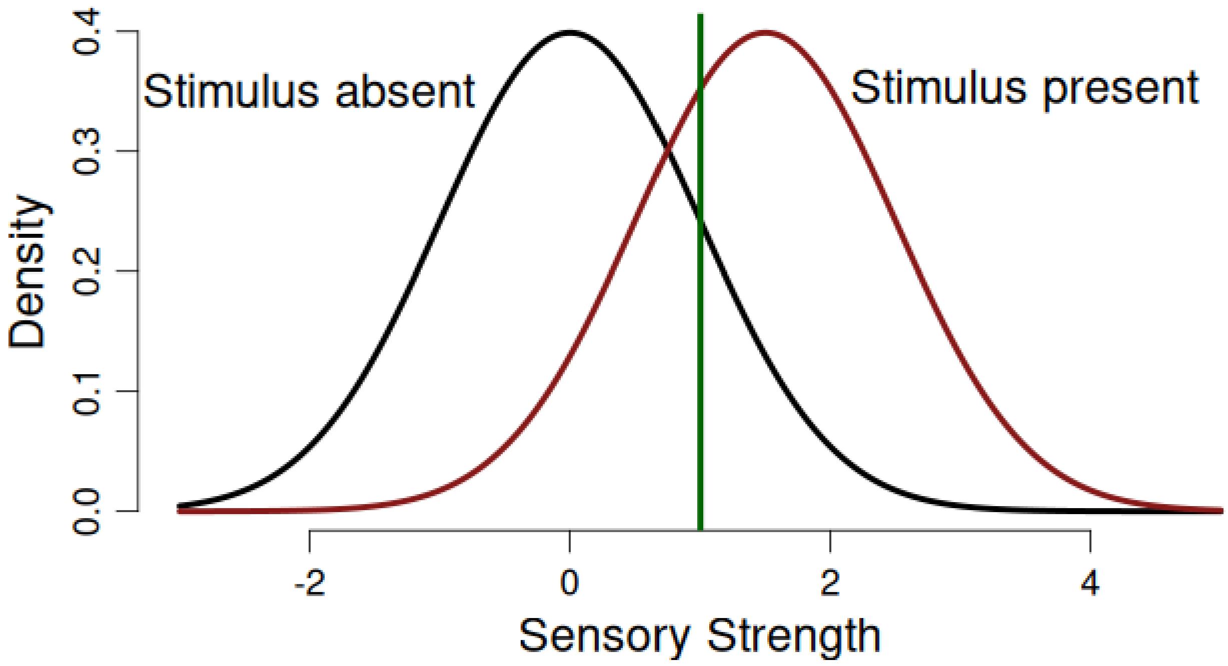
SDT model



$c =$ Criterion, determines the response made.

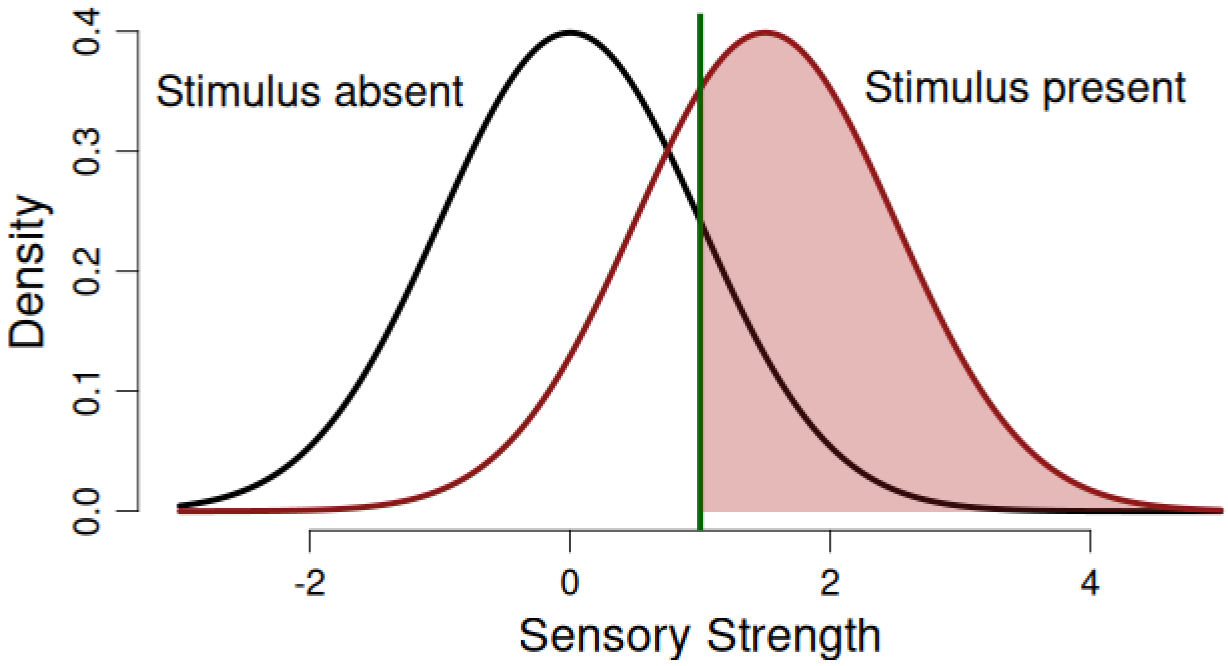
SDT model predictions for H, M, F, C

What corresponds to the probability of hit?



SDT model predictions for H, M, F, C

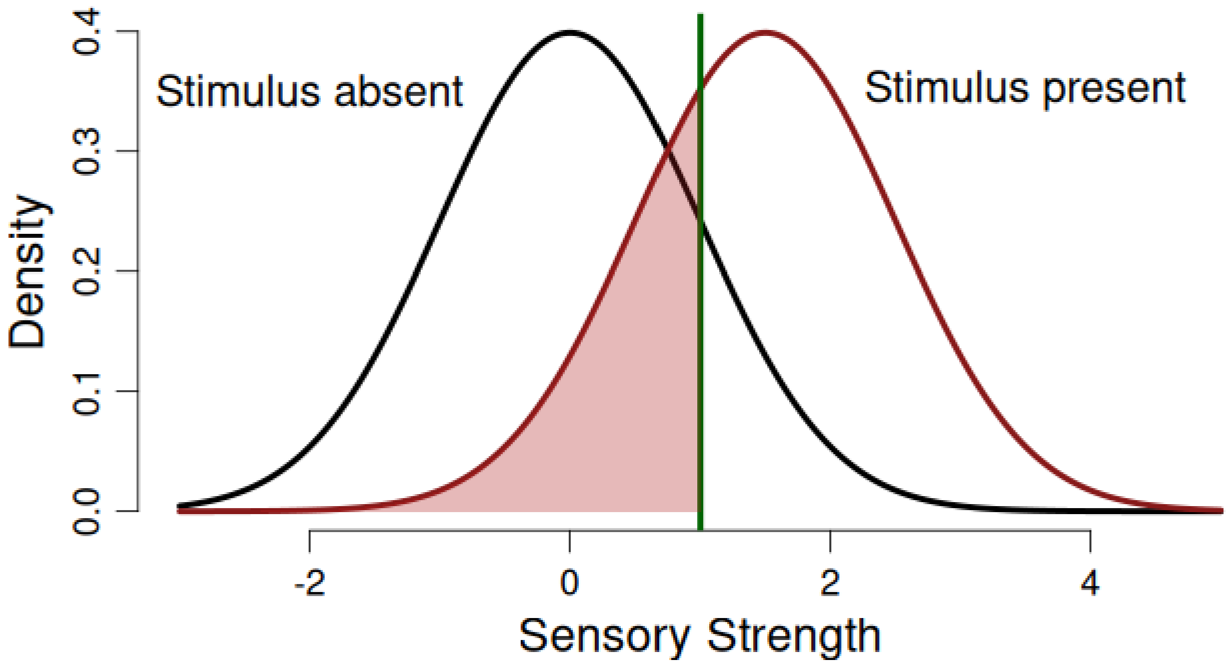
What corresponds to the probability of hit?



Area under the curve!

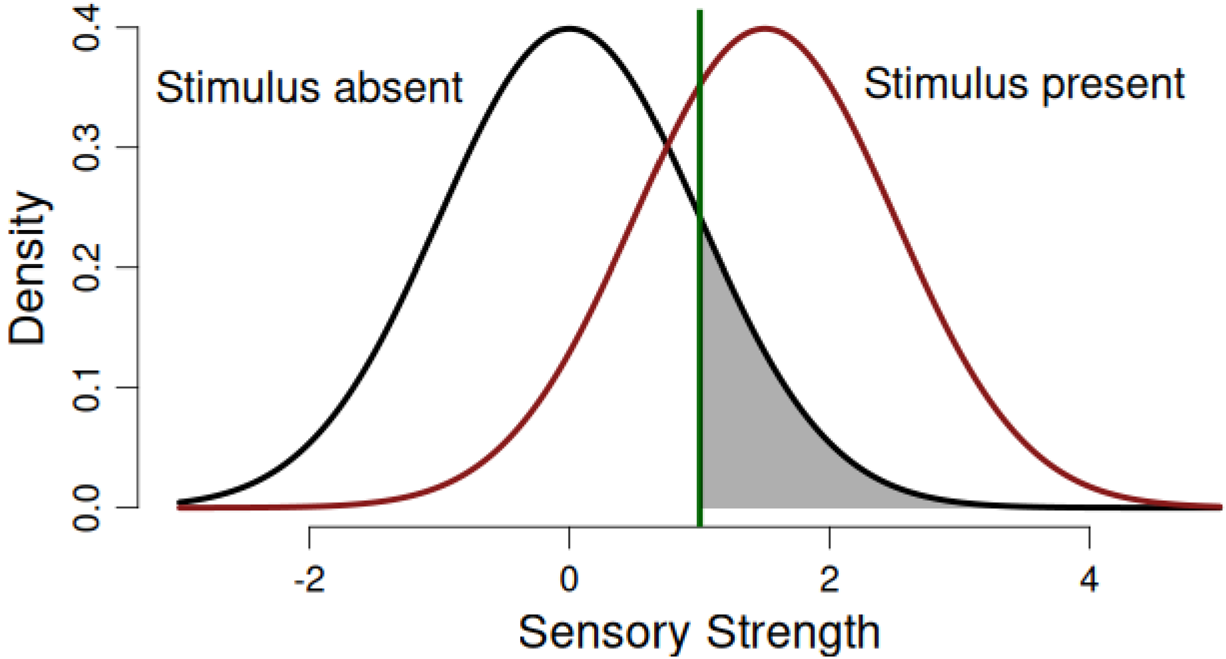
SDT model predictions for H, M, F, C

What corresponds to the probability of miss?



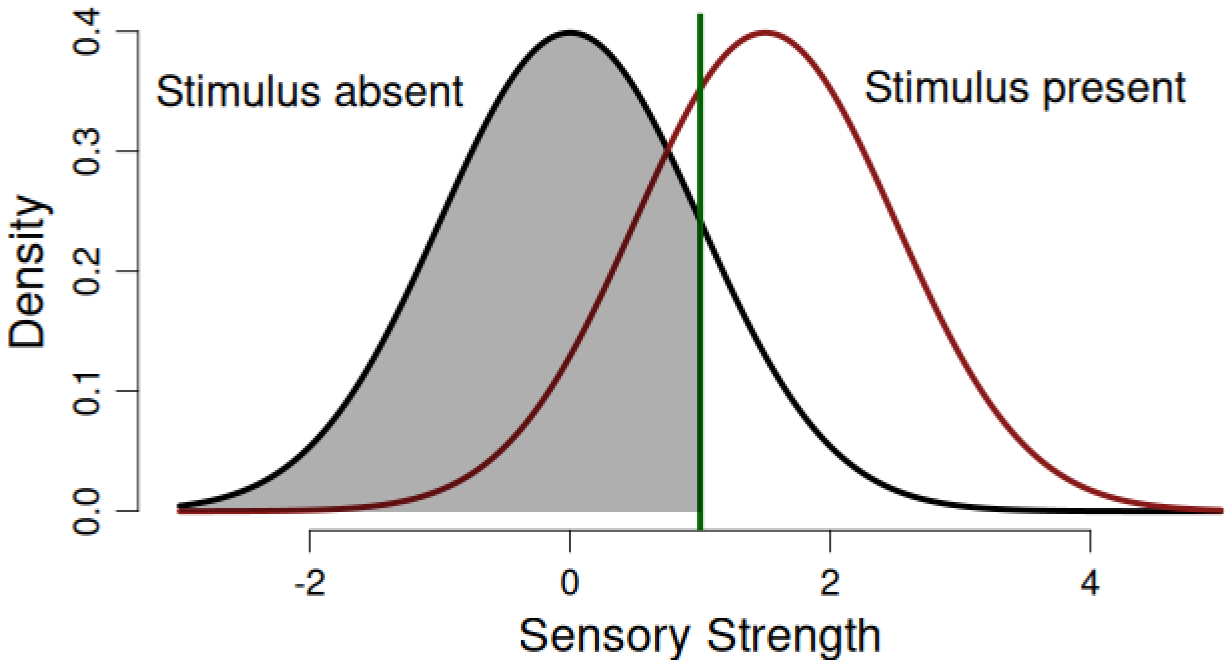
SDT model predictions for H, M, F, C

What corresponds to the probability of false alarm?

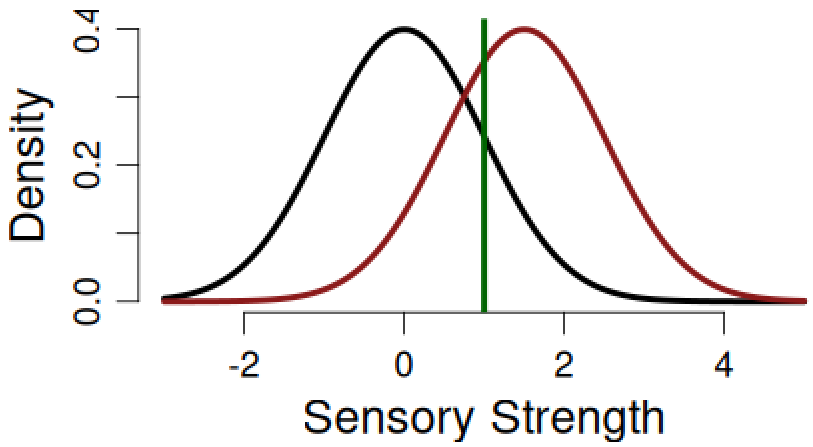


SDT model predictions for H, M, F, C

What corresponds to the probability of correct rejection?



SDT model equations



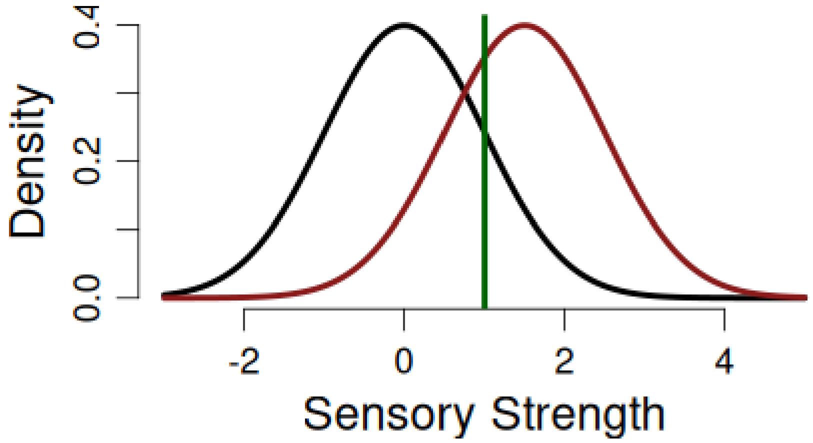

```r
c <- 0.5; dprime = 1

pnorm(c, mean = dprime, sd = 1)
```

```
## [1] 0.3085375
```

```r
pnorm(c - dprime, mean = 0, sd = 1)
```

```
## [1] 0.3085375
```

SDT model equations



Equations for hits, p_1, and false alarms, p_2:

$$p_1 = p(\text{"Signal"} \mid \text{Signal}) = 1 - \Phi(c - d'),$$
$$p_2 = p(\text{"Signal"} \mid \text{Noise}) = 1 - \Phi(c)$$

$\Phi =$ CDF of a standard normal.

SDT in `brms`

Based on the tutorial by Matti Vuorre: https://vuorre.com/posts/sdt-regression/index.html

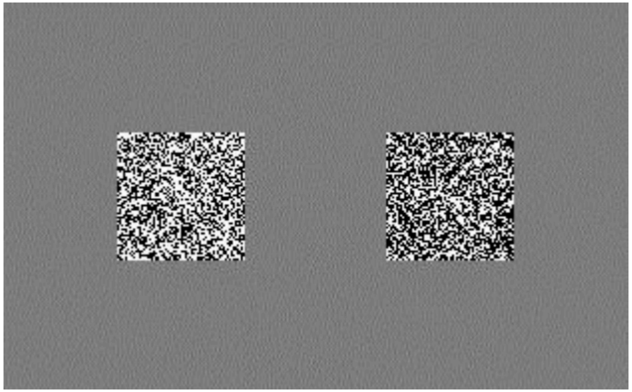
- 48 participants studied a list of 200 words
- Participants were tested with 200 new and 200 old words.
- 6-point Likert item (1: “sure new”, 2: “maybe new”, 3: “guess new”, 4: “guess old”, 5: “maybe old”, 6: “sure old”).
- For now, we bin the rating responses to binary “new” (1-3) and “old” (4-6) responses.

```
data(roc6, package = "MPTinR")
```

```r
fit_glmm <- brm(
  response ~ stimulus + (stimulus | pid),
  family = bernoulli(link = "probit"),
  data = dat
)
```

Drift-Diffusion Model

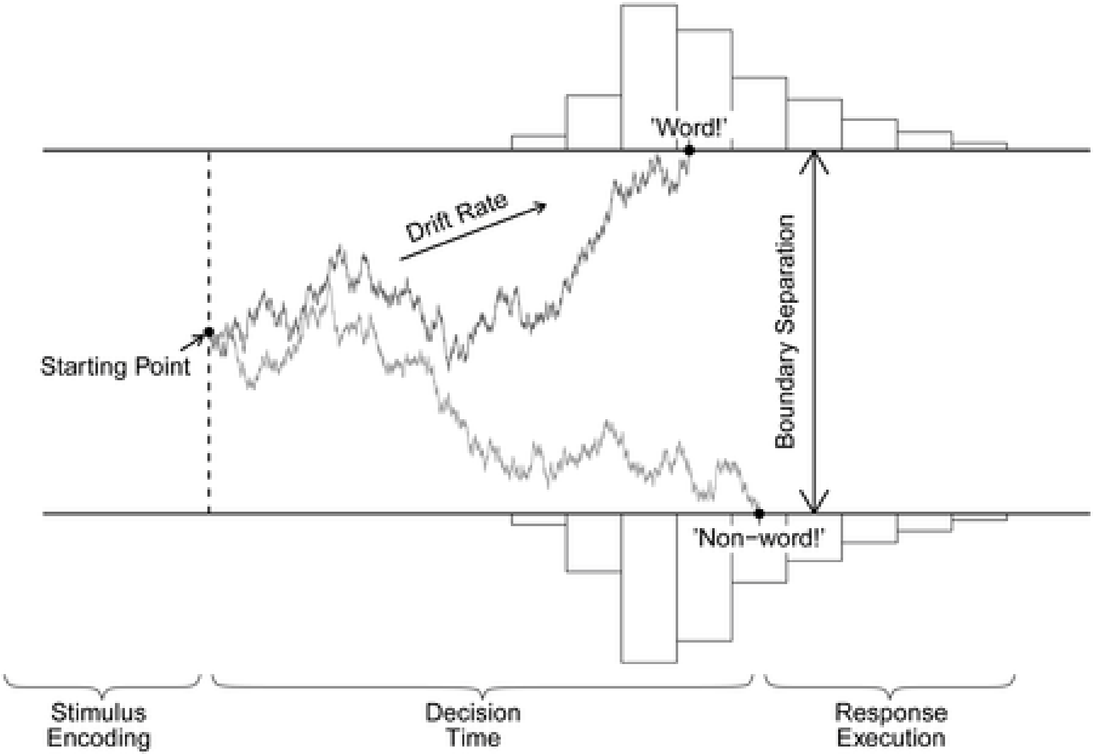
Drift-Diffusion Model



Example: Brightness discrimination task (Ratcliff & Rouder, 1998)

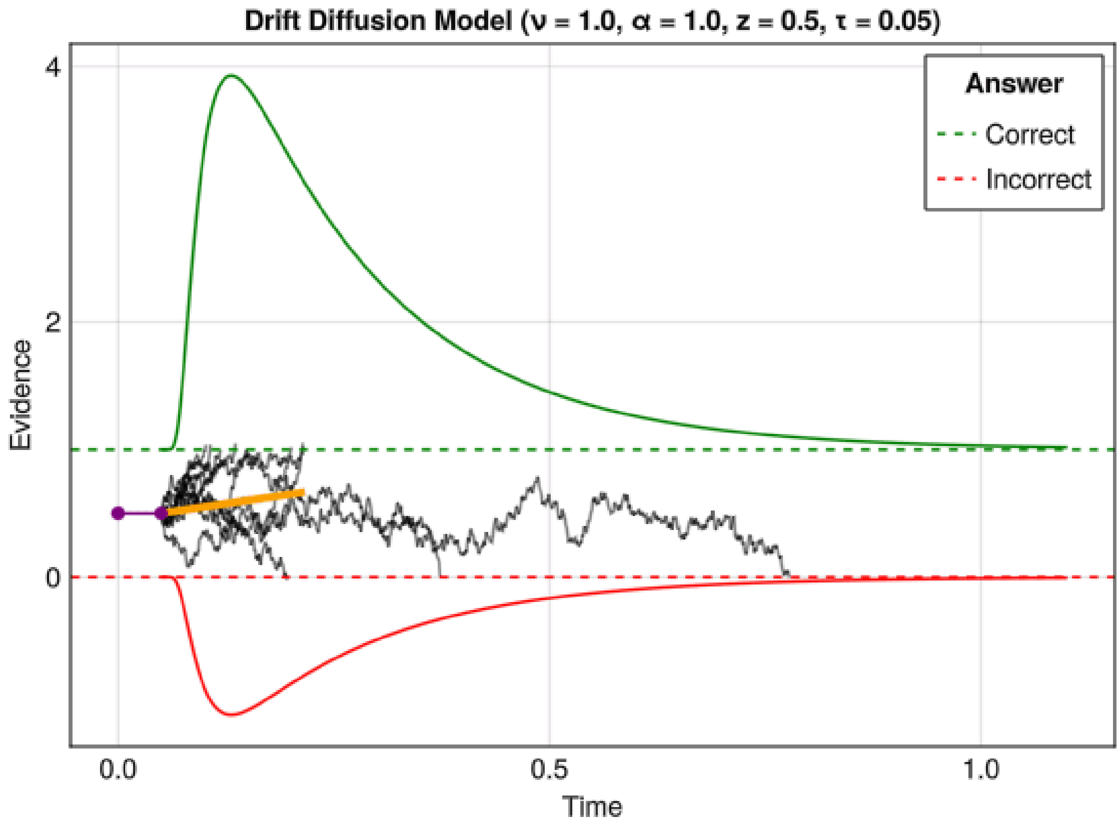
- N = 3, K = 8000
- Manipulated the brightness and response urgency
- Manipulations affect accuracy and response time!

Drift-Diffusion Model



Parameters τ (non-decision time), β (starting point), α (boundary separation), and ν (drift rate).

Drift-Diffusion Model



Changes in τ (non-decision time), β (starting point), α (boundary separation), and ν (drift rate).

DDM in `brms`

Based on the tutorial by Henrik Singmann:
http://singmann.org/wiener-model-analysis-with-brms-part-i/

- Data from Wagenmakers, Ratcliff, Gomez, & McKoon (2008, Experiment 1)
- N=17 participants, lexical decision task: Is the presented string a word or non-word
- Speed or accuracy emphasis instructions in different experimental blocks

```
data(speed_acc, package = "rtdists")
```

DDM in brms

```
formula <- bf(rt | dec(response2) ~ 0 + condition:frequency +
                (0 + condition:frequency|p|id),
               bs ~ 0 + condition + (0 + condition|p|id),
               ndt ~ 0 + condition + (0 + condition|p|id),
               bias ~ 0 + condition + (0 + condition|p|id))
```

```
fit_wiener <- brm(formula,
                  data = speed_acc,
                  family = wiener(link_bs = "identity",
                                  link_ndt = "identity",
                                  link_bias = "identity"),
                  prior = prior, inits = initfun,
                  iter = 1000, warmup = 500,
                  chains = 4, cores = 4,
                  control = list(max_treedepth = 15))
```

Further Topics

Power Analysis for Bayesians?

- What is the goal of power analysis?
- Problem for Bayesians?

Optional stopping: No problem for Bayesians

Jeffrey N. Rouder

Published online: 22 March 2014

Abstract Optional stopping refers to the practice of peeking at data and then, based on the results, deciding whether or not to continue an experiment. In the context of ordinary significance-testing analysis, optional stopping is discouraged, because it necessarily leads to increased type I error rates over nominal values. This article addresses whether optional stopping is problematic for Bayesian inference with

common focus is now on identifying practices that violate the assumptions of our methods, and examples include peeking at the results to decide whether to collect more data (called *optional stopping*) and making additional inferential comparisons that were not considered before data collection. These questionable practices go under the moniker of *p-hacking*, and a remedy for the crisis is to avoid these bad

Power Analysis for Bayesians?

- Bayesian Design Analysis
- Goal: Making studies more informative

A tutorial on Bayes Factor Design Analysis using an informed prior

Angelika M. Stefan[1] · Quentin F. Gronau[1] · Felix D. Schönbrodt[2] · Eric-Jan Wagenmakers[1]

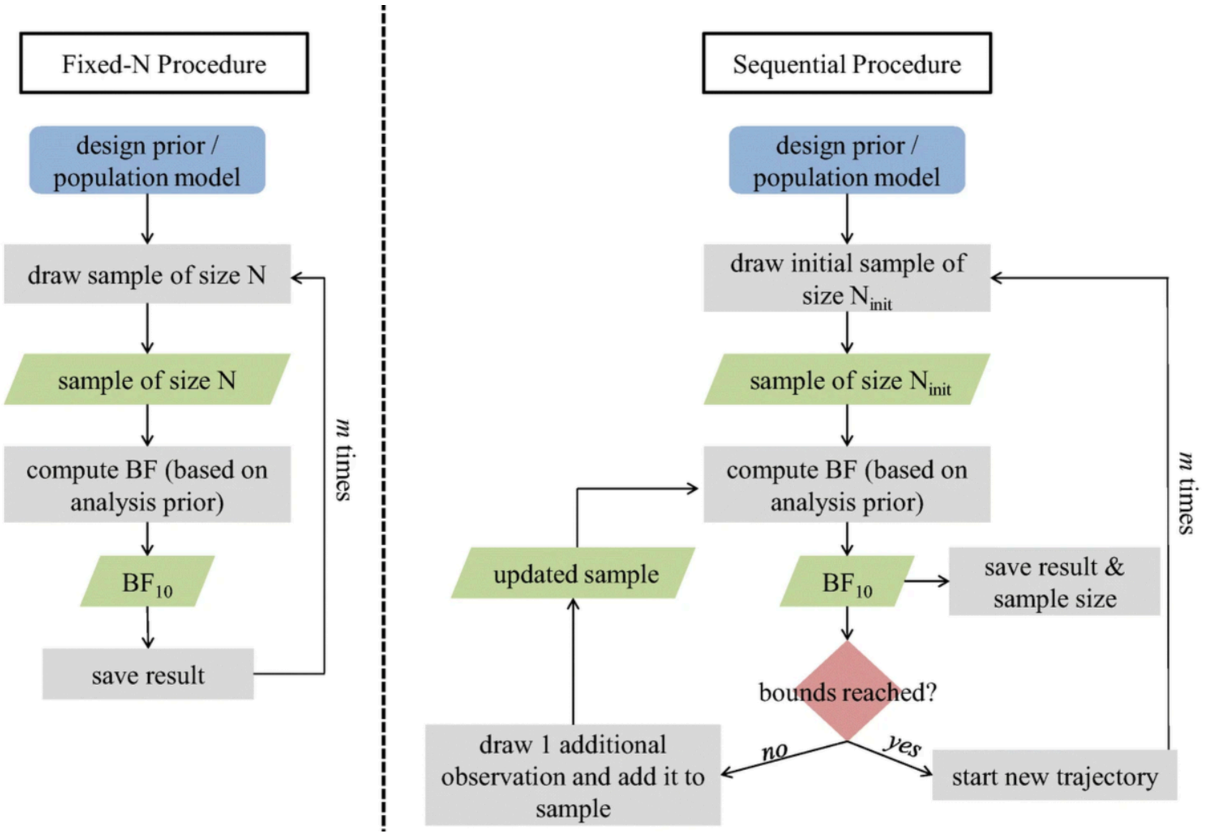
Published online: 4 February 2019

Abstract
Well-designed experiments are likely to yield compelling evidence with efficient sample sizes. Bayes Factor Design Analysis (BFDA) is a recently developed methodology that allows researchers to balance the informativeness and efficiency of their experiment (Schönbrodt & Wagenmakers, *Psychonomic Bulletin & Review*, *25*(1), 128–142 2018). With BFDA, researchers can control the rate of misleading evidence but, in addition, they can plan for a target strength of evidence. BFDA can be applied to fixed-N and sequential designs. In this tutorial paper, we provide an introduction to BFDA and analyze how the use of informed prior distributions affects the results of the BFDA. We also present a user-friendly web-based BFDA application that allows researchers to conduct BFDAs with ease. Two practical examples highlight how researchers can use a BFDA to plan for informative and efficient research designs.

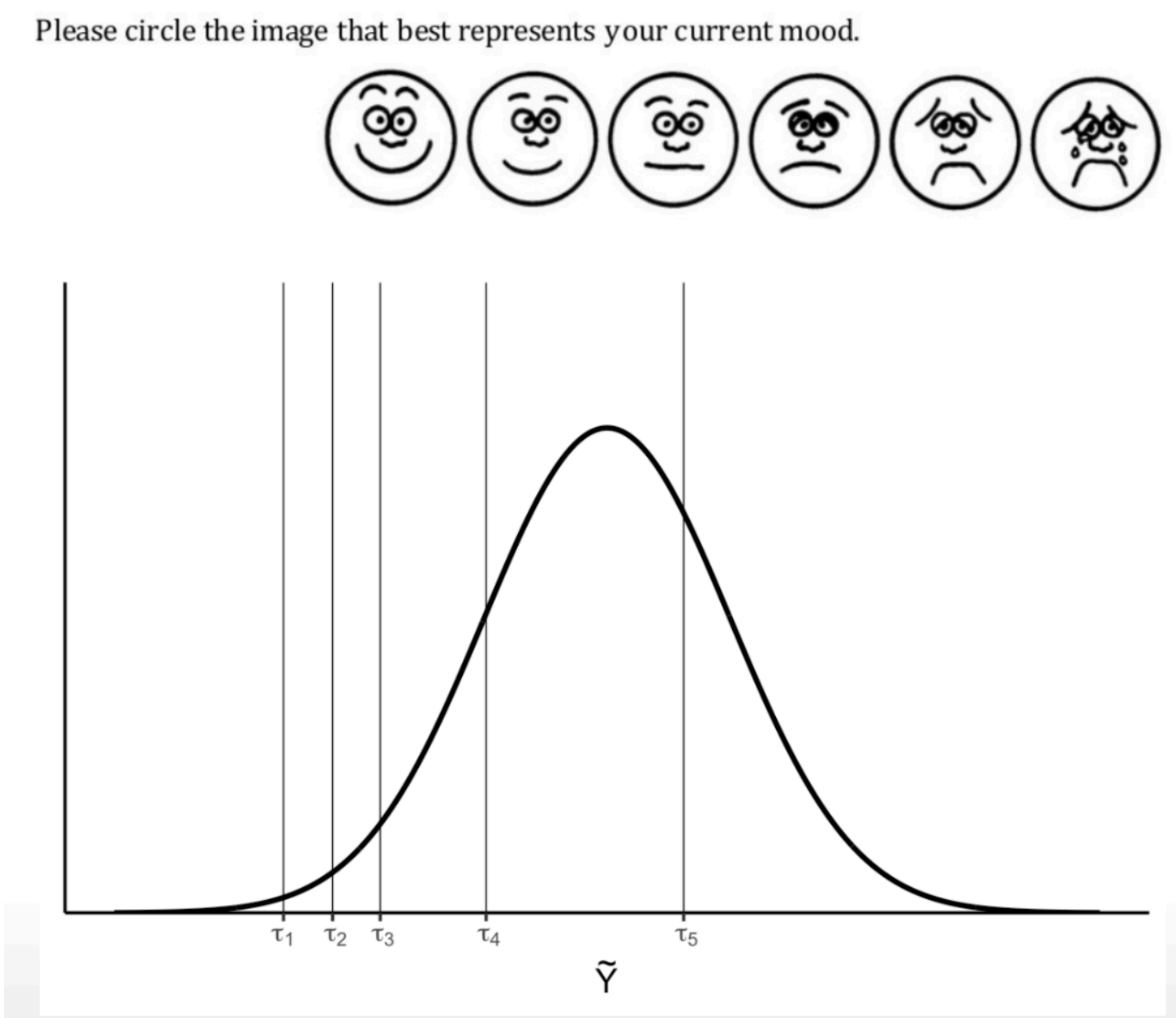
Keywords Sample size · Design analysis · Bayes factor · Power analysis · Statistical evidence

Power Analysis for Bayesians?

- Bayesian Design Analysis
- Goal: Making studies more informative



Ordinal Models in `brms`



Ordinal Models in `brms`

Tutorial paper and other materials are available here:

https://vuorre.com/brms-workshop/posts/ordinal/

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